# Aligned Mathematical Structures

Source-backed grouped formula sample

## 1 Expressions

**Expression 1.** The following expression is taken from a source-backed formula corpus.

$$\left( \begin{array}{c} p_0' \\ p_1' \end{array} \right) = \frac{1}{\sqrt{1-B_{01}^2}} \left( \begin{array}{cc} 1 & 0 \\ -B_{01} & 1-B_{01}^2 \end{array} \right) \left( \begin{array}{c} p_0 - B_{01} p_1 \\ p_1 \end{array} \right) = \left( \begin{array}{c} \gamma p_0 - \gamma v p_1 \\ -v \gamma p_0 + \gamma p_1 \end{array} \right)$$

**Expression 2.** The following expression is taken from a source-backed formula corpus.

$$\begin{array}{lcllcl} g_{rr} & = & \frac{l^2}{r^2} + O(r^{-4}) & g_{tt} & = & -\frac{r^2}{l^2} + O(1) \\ beginalign*2mm]g_{tr} & = & O(r^{-3}) & g_{\varphi\varphi} & = & r^2 + O(1) \\ beginalign*1mm]g_{\varphi r} & = & O(r^{-3}) & g_{t\varphi} & = & O(1) \end{array}$$

**Expression 3.** The following expression is taken from a source-backed formula corpus.

$$(\Gamma^0)^{\underline{\alpha}}{}_{\underline{\beta}} = \left( \begin{array}{cc} 0 & -1 \\ 1 & 0 \end{array} \right), \quad (\Gamma^1)^{\underline{\alpha}}{}_{\underline{\beta}} = \left( \begin{array}{cc} -1 & 0 \\ 0 & 1 \end{array} \right), \quad \Gamma^2 = \Gamma^0 \Gamma^1 = \left( \begin{array}{cc} 0 & -1 \\ -1 & 0 \end{array} \right).$$

**Expression 4.** The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} A_N &= A_1(w_{N-1})\, A_{N-1} = \prod_{i=0}^{N-1} A_1(w_{N-i-1}) \\ D_N &= D_1(w_{N-1})\, D_{N-1} = \prod_{i=0}^{N-1} D_1(w_{N-i-1}) \\ beginalign*0.3em]C_N &= C_1(w_{N-1})\, A_{N-1} + D_1(w_{N-1})\, C_{N-1} \\ &= \sum_{i=0}^{N-1} D_i(w_{N-1-i})\, C_1(w_{N-1-i})\, A_{N-1-i}. \end{aligned}$$

**Expression 5.** The following expression is taken from a source-backed formula corpus.

$$
\begin{aligned}
R^A_{BCD} &= \tilde{R}^A_{BCD}, \\
R^A_{iBj} &= -a_k D_B \partial^A a_l \eta^{kl}_{ij}, \\
R^i_{AjB} &= -a^k D_B \partial_A a_l \eta^{li}_{kj}, \\
R^i_{jkl} &= -a^m \partial^A a_n a_r \partial_A a_s (\eta^{ni}_{mk}\eta^{rs}_{jl} - \eta^{ni}_{ml}\eta^{rs}_{jk}).
\end{aligned}
$$